Energi365 Project Report
CO_2 Storage Simulation Based on SPE-CSP 11B

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Abstract

This report presents a reservoir simulation study of geological CO_2 storage based on the 11th SPE Comparative Solution Project benchmark case SPE11B [1]. I made three simulation variants using OPM Flow: a base case (Task 2) using pore-volume multipliers, a variant (Task 5) with numerical aquifers, and a sensitivity case (Task 6) with increased vertical permeability PERMZ from 0.1 to 0.5 times PERMX. The model covers a 2D cross-section of 8400 m horizontally and 1200 m vertically, with seven geological facies and two injection wells over a 50-year injection period followed by 950 years of monitoring. Results show CO_2 rising buoyantly and accumulating beneath the low permeability seal facies, with progressive dissolving Co_2 into formation water over time. The aquifer boundary condition changes CO_2 migration behaviour, while increased PERMZ leads to a greater seal breach.

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1 Introduction

This project concerns reservoir simulation of geological carbon storage (GCS), a key technology for reducing CO_2 emissions. Safe long-term storage in deep saline aquifers requires simulation to predict CO_2 migration, pressure evolution, and trapping over timescales of hundreds to thousands of years. The 11th SPE Comparative Solution Project (SPE11B) provides a realistic representative of Norwegian Continental Shelf storage conditions, previously studied by several international research groups [2].

SPE11B is a 2D vertical cross-section of 8400 m $\times$ 1200 m, containing seven geological facies, two CO_2 injection wells, and a 1000-year simulation horizon. A key feature is a permeable fault zone (Facies 3) connecting the lower and upper reservoir, providing a migration pathway toward the secondary storage region (Box B).

The objective was to construct the simulation input from the benchmark description, run the simulation using OPM Flow, and analyse the results in terms of CO_2 distribution, pressure evolution, and trapping.

2 Methods

2.1 Simulation Data

All simulations were run using OPM Flow 2025.10 with the `CO2STORE` and `DISGASW` options activated, enabling CO_2 fluid properties and dissolved CO_2 tracking. Include files provided as part of the project resources were used for grid geometry (`DX.INC`), permeability (`PERMX.INC`), porosity (`PORO.INC`), saturation tables (`SGWFN.INC`), boundary volumes (`PVBOUNDARIES.INC`), and the saturation and fluid regions (`SATNUM.INC`, `FIPNUM.INC`). PERMZ was set to 0.1 times PERMX implementing the 10:1 anisotropy from the benchmark. The reservoir was initialised using `EQUIL` at 2900 m depth with a reference pressure of 300 bar, and a uniform temperature of 70°C was set via `RTEMP`. INJ1 injects CO_2 at 1600 Sm^3/day from year 0 to year 50, with INJ2 joining at year 25 also at 1600 Sm^3/day, both shutting in at year 50 followed by 950 years of monitoring. Diffusion coefficients `DIFFAWAT` and `DIFFAGAS` were included, but `DIFFUSE` was not activated in Task 2 and Task 6 due to a simulation abort, but was successfully activated in Task 5 only. As a result, the RSW finger patterns in Task 2 and Task 6 are driven by numerical diffusion and are internally grid dependent rather than physically.

For clean figures all summary results were extracted from OPM Flow's CSV output files and post-processed in Python by summing the relevant FIPNUM regions, for example Box A as regions 2, 4 and 5 and Box B as regions 3 and 6, before plotting all three tasks together for comparison.

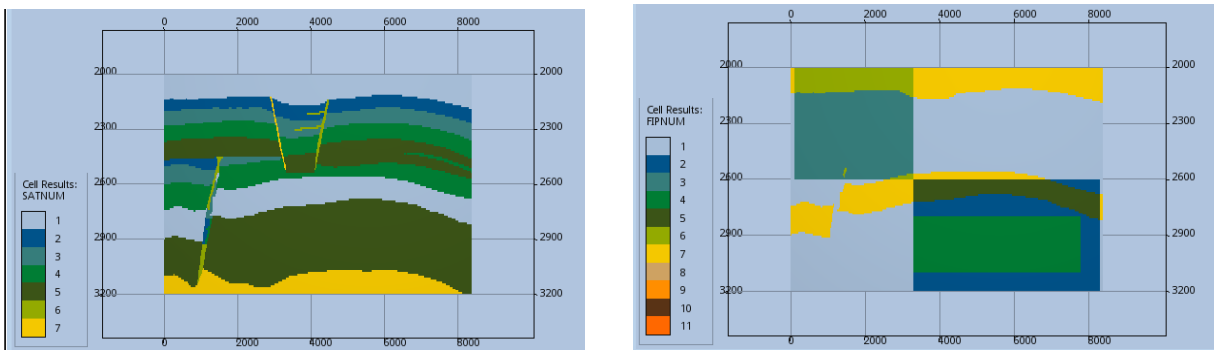
2.2 Grid, Rock Properties and Saturation Functions

The model grid has dimensions 842 $\times$ 1 $\times$ 120 cells (101,040 total), with uniform DZ of 10 m giving 1200 m vertical extent and the top placed at 2000 m depth. PERMZ is derived from PERMX by a multiplier of 0.1, implementing the 10:1 anisotropy from the benchmark (Eq. 3.10 in [1]).

The saturation function tables were generated independently using a spreadsheet following the Brooks–Corey formulation (Eqs. 2.6–2.9 and 3.11 in [1]), with seven separate tables built one per facies. The capillary entry pressure was calculated using Leverett J-scaling: $p_{\text{entry}} = \sqrt{\frac{\phi}{k_x}} \cdot 6.12 \times 10^{-3}$ N/m Relative permeabilities were computed from $k_{r,\alpha} = (s_{\alpha,n})^{1.5}$ and capillary pressure was capped using the error-function with $p_{\text{cap,max}} = 300$ bar.

2.3 Reporting Regions and Initial Conditions

The `FIPNUM` array defines 11 reporting regions, shown together with the SATNUM map in Figure 1. Box A (primary storage) corresponds to regions 2, 4 and 5, Box B (secondary storage) to regions 3 and 6, the seal within Box A to region 5, the seal within Box B to region 6, and the total seal to region 7. From the SATNUM map the reservouir structure is also visible. Looking at the saturation table together with the SGAS results in Resinsight I saw that Facies 1 is the low-permeability seal that traps CO_2 beneath it, Facies 7 is fully impermeable, and Facies 2 through 6 are more permeable layers where the CO_2 flows more freely. The diagonal fault (Facies 3) acts as the main migration pathway for CO_2 from Well 1 toward Box B, connecting the lower reservoir to the upper storage units. Since Facies 1 has a high capillary entry pressure, CO_2 preferentially migrates through the fault rather than directly through the seal.



(a) SATNUM – seven geological facies

(b) FIPNUM – eleven reporting regions

Figure 1: SATNUM and FIPNUM property maps as visualised in ResInsight.

2.4 Task Variants

Table 1 summarises the three simulation variants. Task 2 uses pore-volume multipliers via `PVBOUNDARIES.INC` as the boundary condition. Task 5 replaces this with two numerical aquifers using `AQUNUM` and `AQUCON`, connecting to the left boundary (I=2, K=91–110) and right boundary (I=841, K=83–120) at the Facies 5 region, both with Facies 5 properties ($\phi = 0.25$, $k = 1000$ mD) with length 50,000 m. Task 6 is identical to Task 2 except PERMZ is increased from 0.1 to 0.5 times PERMX, reducing the anisotropy ratio from 10:1 to 2:1.

Table 1: Summary of the three simulation variants.

Feature	Task 2 (Base)	Task 5 (Aquifer)	Task 6 (PERMZ)
Boundary volumes	PV multipliers	AQUNUM + AQUCON	PV multipliers
PERMZ multiplier	0.1	0.1	0.5
DIFFUSE active	No	Yes	No

3 Results and Discussion

3.1 Injection Schedule

Figure 2 shows WGIR, WGIT and WBHP per well for Task 2, confirming both wells injected at 1600 $Sm3^3$ 3/day each as specified. I also looked at FGIR which is the total field injection rate — it starts at 1600 $Sm3^3$ 3/day when only INJ1 is active and doubles to 3200 $Sm3^3$ 3/day at year 25 when INJ2 starts. I also noticed that FGIP and FGIT remain close to identical throughout the simulation, meaning that all the injected CO_2 stays within the domain and is not escaping

to the boundaries. The bottom hole pressure at INJ1 rises earlier and stabilises at a higher value than INJ2, because INJ1 being placed deeper in the reservoir where pressure of course is higher.

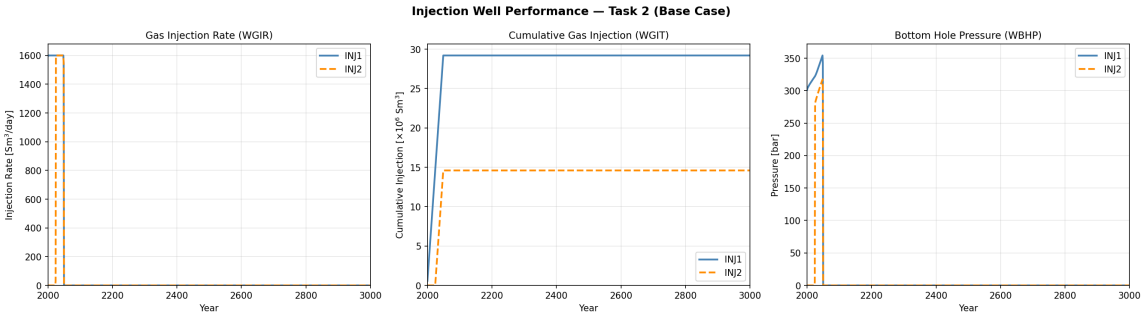
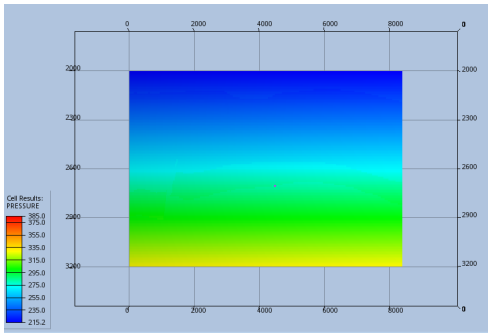


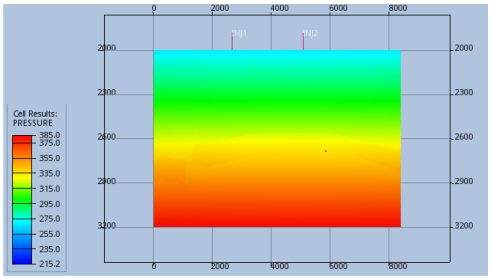
Figure 2: Gas injection rate (WGIR), cumulative injection (WGIT) and bottom hole pressure (WBHP) per well for Task 2.

3.2 Pressure Evolution

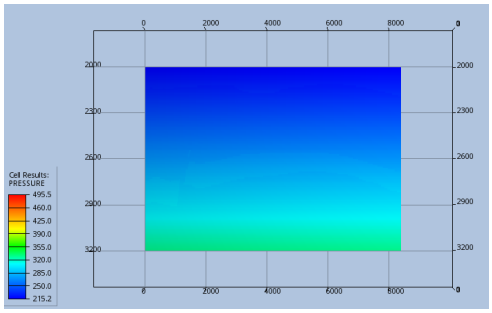
Figure 3 shows the pressure distribution at year 0 and year 1000 for Task 2 and Task 5. We can see the initial hydrostatic gradient in both, with higher pressure at the bottom due to the weight of the water column above. Task 5 shows a slightly lower initial pressure overall, likely because the numerical aquifers connect only to a limited interval of boundary cells compared to the distributed pore-volume multipliers in Task 2 which acts on cells across all the permeable facies, providing more initial volume support. By year 1000, Task 5 shows substantially higher pressure than Task 2. The aquifer actively pushes water inward from both boundaries rather than simply absorbing pressure as the PV multipliers do, leading to a larger pressure buildup inside the domain over the simulation period.



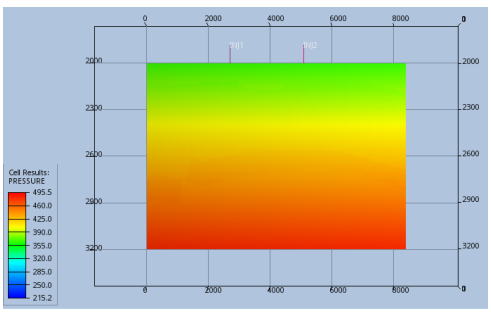
(a) Task 2 – Year 0



(b) Task 2 – Year 1000



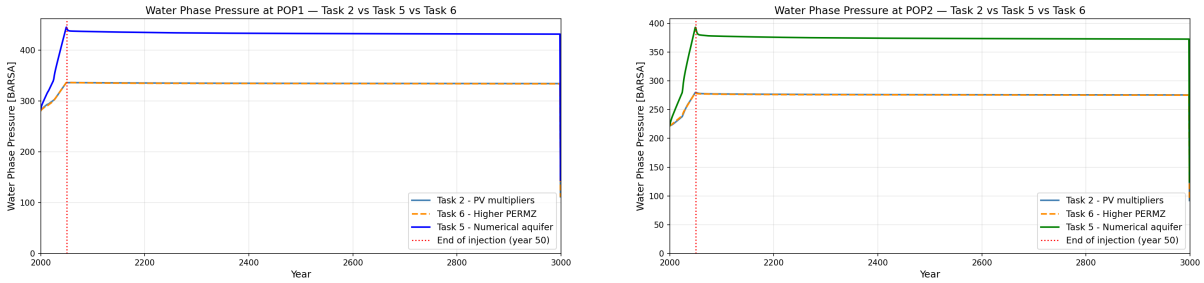
(c) Task 5 – Year 0



(d) Task 5 – Year 1000

Figure 3: Spatial pressure distribution at year 0 and year 1000 for Task 2 and Task 5.

Figure 4 shows water phase pressure at POP1 (mid-reservoir, Facies 5) and POP2 (near seal, Facies 1) for all three tasks. Both points show a pressure increase up to year 50 followed by stabilization through to year 1000, indicating pressure acts as a fully connected system through the reservoir with no measurable dampening from the different pemerabilities Facies 1 or 5. POP1 remains at a higher absolute pressure than POP2 throughout due to its greater depth and corresponding hydrostatic head. Task 5 shows higher pressure at both points due to active water influx from the numerical aquifer, while Task 6 follows Task 2 identically, confirming the PERMZ change has no effect on the overall pressure field.



(a) POP1 – mid-reservoir (Facies 5)

(b) POP2 – near seal (Facies 1)

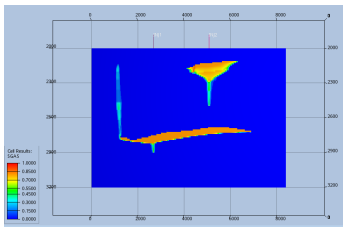
Figure 4: Water phase pressure at POP1 and POP2 for all three tasks over 1000 years.

3.3 CO_2 Distribution

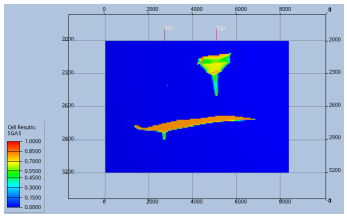
Figures 5 and 6 show SGAS and RSW for all three tasks at year 50, 200 and 1000. SGAS shows where free phase CO_2 is present, with orange and red indicating free-phase CO_2 and blue indicating water saturated regions. RSW shows where CO_2 has dissolved into the brine, with the downward fingers resulting from convective mixing as dissolved CO_2 increases brine density causing it to sink while unsaturated water rises to replace it.

In Task 2 at year 50, CO_2 from Well 1 rises vertically before spreading horizontally beneath the seal and migrating up the fault toward Box B, while Well 2 spreads horizontally beneath the top seal. By year 200 the CO_2 has accumulated in three main areas: a large plume under the primary seal in Box A, a smaller plume where Well 2 reaches the top seal, and a even smaller plume that has migrated through the fault into Box B. We can note that this seems consistent, as Well 1 has injected twice the total amount of CO_2 compared to Well 2, and only a fraction of the CO_2 from Well 1 migrates through the fault up into Box B. By year 1000 free-phase CO_2 has reduced further as dissolution takes over. Task 5 shows CO_2 confined to the lower reservoir with less fault migration, as the aquifer pushes water inward from both sides blocking the migration pathway. Task 6 shows a more CO_2 at box Bat year 50 due to the higher PERMZ, and by year 200 the free-phase CO_2 has already mostly dissolved, so the higher PERMZ accelerates the transition to dissolved CO_2.

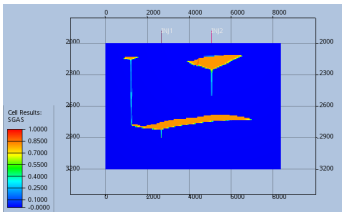
The RSW fingering develops more extensively over time in all tasks. Task 6 shows more vigorous and earlier finger development than Task 2, consistent with the higher PERMZ making it easier for dense CO_2-rich water to sink. As discussed in section 2.1, the finger patterns in Task 2 and Task 6 are driven by numerical diffusion rather than physical molecular diffusion since `DIFFUSE` was not active in these tasks. Task 5 includes physical diffusion and produces more physically realistic finger patterns, with more developed fingers extending downward in a structured way beneath the plumes, consistent with convective mixing.



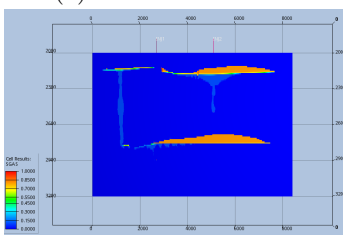
(a) Task 2 – Year 50



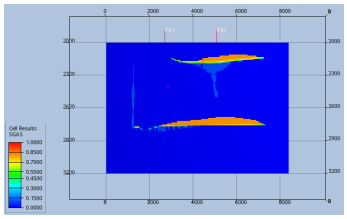
(b) Task 5 – Year 50



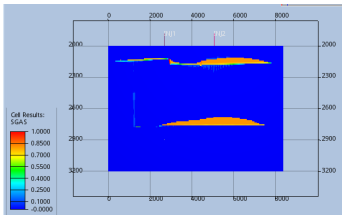
(c) Task 6 – Year 50



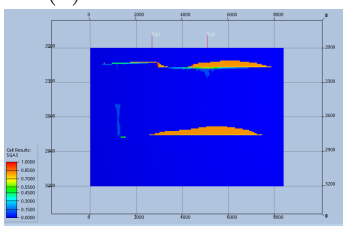
(d) Task 2 – Year 200



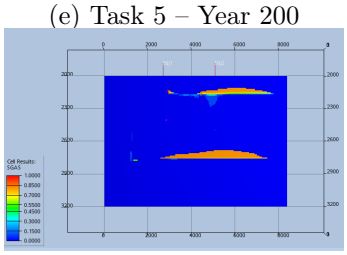
(e) Task 5 – Year 200



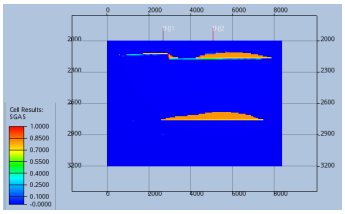
(f) Task 6 – Year 200



(g) Task 2 – Year 1000

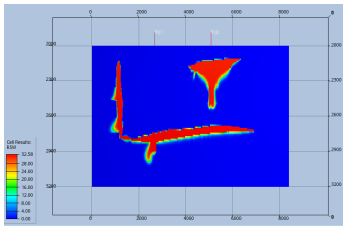


(h) Task 5 – Year 1000

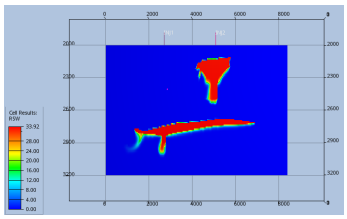


(i) Task 6 – Year 1000

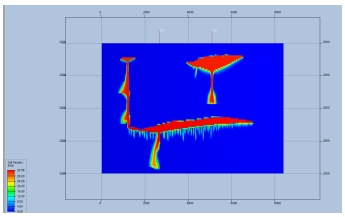
Figure 5: Gas saturation (SGAS) for Task 2, Task 5 and Task 6 at year 50, 200 and 1000.



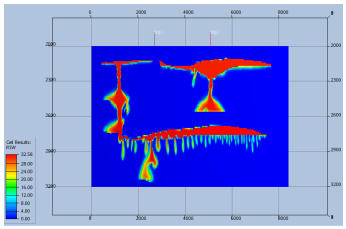
(a) Task 2 – Year 50



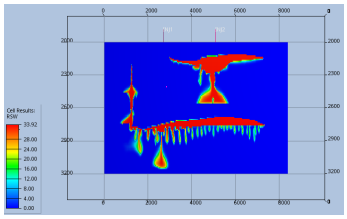
(b) Task 5 – Year 50



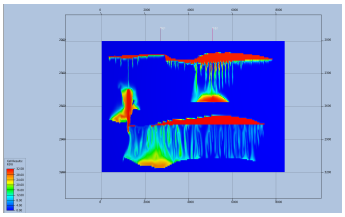
(c) Task 6 – Year 50



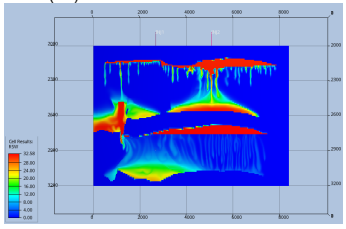
(d) Task 2 – Year 200



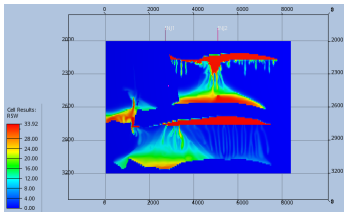
(e) Task 5 – Year 200



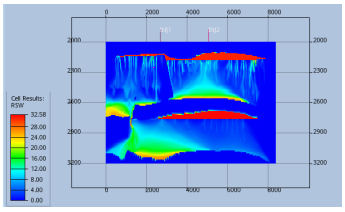
(f) Task 6 – Year 200



(g) Task 2 – Year 1000



(h) Task 5 – Year 1000



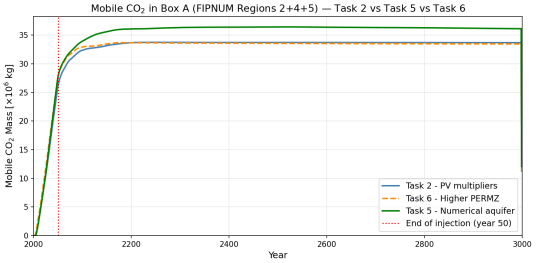
(i) Task 6 – Year 1000

Figure 6: Dissolved CO_2 in the water phase (RSW) for Task 2, Task 5 and Task 6 at year 50, 200 and 1000.

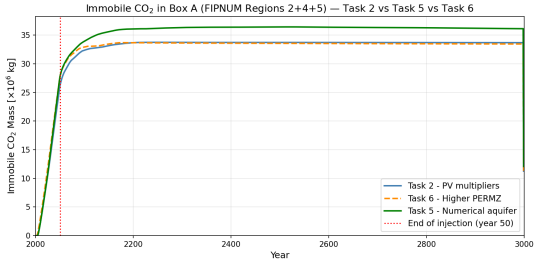
3.4 CO_2 in Place by Region

Figures 7 and 8 show mobile, immobile and dissolved CO_2 in Box A and Box B for all three tasks. I looked at the (CSV-file) and noticed that RGMMO and RGMUS are numerically identical for all three tasks in both boxes, which could be because the grid CO_2 saturations is close to the residual saturation threshold, causing the simulator to assign the same mass to both categories rather than separating them.

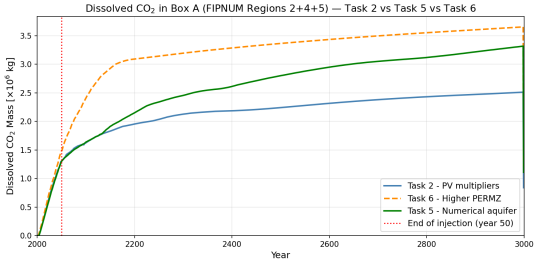
For mobile and immobile CO_2 in Box A, Task 5 shows higher values than Task 2 and Task 6, that remain very similar to each other. The aquifer pushes water inward from both sides, preventing CO_2 from migrating through the fault and instead is concentrating it in Box A. For dissolved CO_2 in Box A, Task 6 shows the highest values throughout, as the higher PERMZ allows CO_2 to reach the brine contact faster and dissolve more quickly. The denser CO_2-saturated brine then sinks but is retained within Box A by the impermeable Facies 7 at the bottom, keeping the dissolved amount of CO_2 high. Task 2 shows the lowest dissolved CO_2 in Box A, as a fraction of its CO_2 migrates through the fault into Box B reducing what remains in Box A, and it also dissolves more slowly than Task 6 due to the lower vertical permeability. Task 5 falls between the two, with higher dissolved CO_2 than Task 2 due to the greater CO_2 concentration kept in Box A by the aquifer, despite dissolving at a likewise rate.



(a) Mobile (RGMMO)



(b) Immobile (RGMUS)



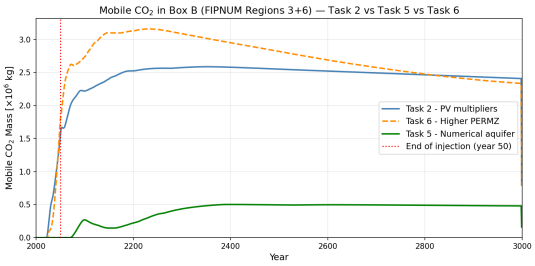
(c) Dissolved (RGMDS)

Figure 7: Mobile, immobile and dissolved CO_2 in Box A (FIPNUM 2+4+5) for all three tasks.

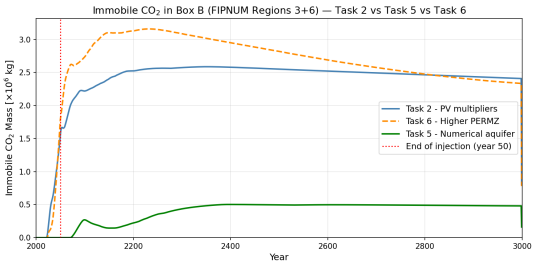
The differences between tasks are more visible in Box B than in Box A. Task 5 shows much lower values across all three categories because the aquifer pressure prevents CO_2 from reaching the secondary storage unit through the fault. Task 6 peaks higher than Task 2 for both mobile and immobile CO_2 in Box B during the early injection period, consistent with the higher vertical permeability driving more CO_2 upward and into Box B faster. But, once injection stops Task 6 drops below Task 2 and continues to decline toward year 1000, as the lower anisotropy allows CO_2 to dissolve and escape Box B faster, leaving Task 2 with the higher free phase CO_2 at the final observation year.

For dissolved CO_2 in Box B, Task 2 and Task 6 increase at a similar rate during injection, but after year 50 Task 2 continues accumulating dissolved CO_2 within Box B while Task 6 declines a little before gradually increasing towards year 1000. This is as expected with the higher PERMZ

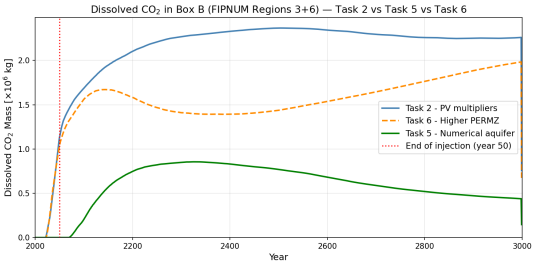
in Task 6 causing the denser CO_2-saturated brine to sink out of Box B faster through convective fingers, so even though Task 6 dissolves CO_2 quickly, it also sinks out of the box more rapidly. Task 2, with its lower vertical permeability, retains the dissolved CO_2 within Box B for longer, leading to higher accumulated values toward year 1000. Task 5 remains lowest throughout as alot of the CO_2 doesnt reach Box B in the first place.



(a) Mobile (RGMMO)



(b) Immobile (RGMUS)



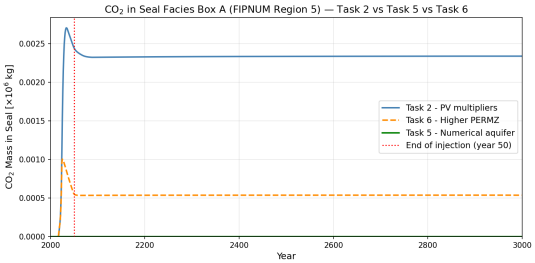
(c) Dissolved (RGMDS)

Figure 8: Mobile, immobile and dissolved CO_2 in Box B (FIPNUM 3+6) for all three tasks.

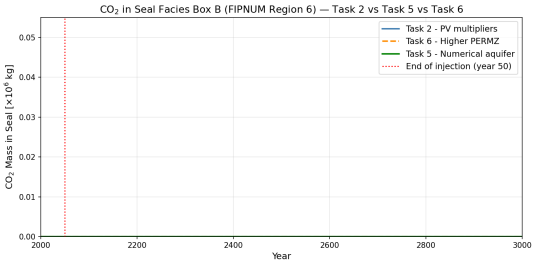
3.5 CO_2 in Seal Facies

Figure 9 shows CO_2 in seal Facies 1 within Box A (region 5), Box B (region 6) and the full domain (region 7). For the seal within Box A, Task 2 shows more CO_2 entering the seal than Task 6, while Task 5 is close to zero. This seems counterintuitive given that Task 5 concentrates more CO_2 beneath the seal in Box A, but could be because the higher water phase pressure from the aquifer influx increasing resistance to capillary entry into Facies 1. For the seal within Box B, all three tasks show values close to zero, which appears inconsistent with the total seal results in region 7. This looks like an error when comparing to the final nordbotten results [2] , but may tell us that most of the seal breach occurs outside the strict boundaries of Box A and Box B, and is therefore not captured by regions 5 and 6 individually.

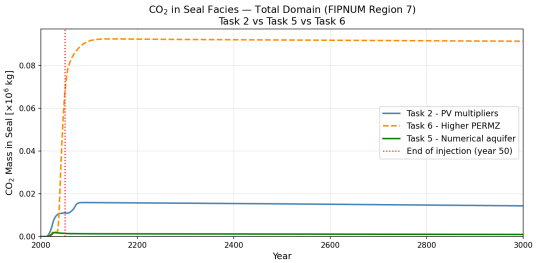
The total seal region 7 reveals the most significant differences between tasks. Task 6 shows substantially more CO_2 entering the seal than Task 2, which is consistent with the higher PERMZ allowing CO_2 to push upward more easily against the capillary entry pressure of Facies 1, resulting in greater seal breach across the full domain. Task 5 shows near-zero CO_2 in the total seal, consistent with the aquifer pressure keeping CO_2 confined to the lower reservoir and away from the seal throughout the simulation.



(a) Seal Box A (region 5)



(b) Seal Box B (region 6)



(c) Total seal (region 7)

Figure 9: CO_2 mass in seal Facies 1 at three spatial levels for all three tasks.

4 Discussion

4.1 Physical Plausibility and Comparison with Benchmark

The simulated CO_2 migration is physically consistent with expectations for this storage scenario. CO_2 rises buoyantly and accumulates beneath the caprock, with dissolution becoming progressively more noticable over the post injection period. Looking at the sparse data plots from Nordbotten et al. (2025) [2] (Figure 10) for SPE11B, my results appear consistent with their findings. The water phase pressure at POP2 reaches a similar value to the range reported across the benchmark groups before stabilising after injection stops. The mobile gaseous CO_2 in Box A also reaches values in a similar range to what is reported across the submissions. For Box B the mobile CO_2 in Task 2 is also within the range of the benchmark results, though on the lower end, which is consistent with the variability between groups spanning a factor of 2–3 as noted by Nordbotten et al. (2025) [2].

One difference is that the CO_2 in seal facies within Box B is zero for all three tasks in the present model, while some of the benchmark submissions report small but non-zero amounts there. This could be a cause to how the boundary conditions are defined for my model. Overall the results appear physically consistent with the benchmark submissions, and the general behaviour of CO_2 rising buoyantly, accumulating beneath the seal, and progressively dissolving into the brine over time is in agreement with what is reported across the participating groups.

4.2 Boundary Conditions and Permeability Sensitivity

Both approaches — PV multipliers and numerical aquifer — are valid implementations of the buffer volume, but represent different assumptions. Pore-volume multipliers simply inflate the pore volume of the boundary cells to add extra volume and prevent pressure buildup. Numerical aquifers is like a huge amount of water at the boundaries that provides a more realistic representation of our model that supports pressure. The higher pressure in Task 5 and the fundamental change in CO_2 migration it produces shows that the choice of boundary condition matters, both for pressure evolution and for where CO_2 ends up in the reservoir.

Increasing PERMZ from 0.1 to 0.5 times PERMX had a noticable effect on the dissolved CO_2 in Box A and shows clear increase in CO_2 entering the seal, which is one of the most interesting results from Task 6 on how it breaches the caprock. The PERMZ change directly targets the vertical flow resistance that controls how quickly CO_2 reaches the seal, and the result confirms that vertical permeability is an important characteristic on seal integrity in the CO_2 storage.

5 Conclusion

This project successfully set up, ran and analysed three variants of the SPE11B CO_2 storage benchmark using OPM Flow. The saturation function tables were generated independently using a spreadsheet following the Brooks–Corey formulation. Task 5 replaces the boundary volumes with numerical aquifers, resulting in higher interior pressure build-up and different CO_2 migration behaviour, with the aquifer preventing CO_2 from reaching Box B. Task 6 increases PERMZ from 0.1 to 0.5 times PERMX, producing an increase in CO_2 entering the total seal facie (Fipnum 7), highlighting the sensitivity of storage security to vertical permeability.

The main practical challenges were generating numerically stable saturation tables, handling a consistent solver convergence error in the simulations during active injection, and the inability to activate `DIFFUSE` in Task 2 and Task 6 due to a conflict with `PVBOUNDARIES.INC`. Given more time, it would be valuable to resolve the diffusion issue for the base case and to compare results more quantitatively against the reference submissions of Nordbotten et al. (2025). Also it would be interesting to modify the capillary pressure curves for individual facies in a meaningful way to observe how CO_2 migration and trapping behaviour changes.

References

[1] Nordbotten, J.M., Fernø, M.A., Flemisch, B., Kovscek, A.R., and Lie, K.A. (2024). *The 11th Society of Petroleum Engineers Comparative Solution Project: Problem Definition*. SPE Journal. https://doi.org/10.2118/218015-PA

[2] Nordbotten, J.M., Fernø, M.A., Flemisch, B., et al. (2025). *Benchmarking CO_2 storage simulations: Results from the 11th Society of Petroleum Engineers Comparative Solution Project*. International Journal of Greenhouse Gas Control 148, 104519. https://doi.org/10.1016/j.ijggc.2025.104519

[3] Society of Petroleum Engineers (2023). *SPE Comparative Solution Project 11*. https://spe.org/en/csp/spe11/

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[5] Nordbotten, J.M. (2025). *Energi365 – Project Work CO_2 Storage*. Project description and resources, University of Bergen.